



WORKSHOP GUIDE

VirtualGlove Controller Enclosure Guide

Print and assemble the UNO Q Case or integrated Controller Dock.

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VirtualGlove now includes two printable homes for the Arduino UNO Q Controller. Both were designed for an Anycubic Kobra 3, ordinary PLA, a 0.4 mm nozzle, and the Arduino USB-C Hub (8 in 1).

The files are intentionally parametric. Print the small fit tests first, record what changed, and adjust the clearances before committing to a long print.

Choose your case



UNO Q Case exterior

UNO Q Case

Choose the compact case when the Arduino hub will sit elsewhere in the arcade panel. The case protects the UNO Q, exposes only its USB-C connection, keeps the Matrix visible, and leaves airflow around the board.

- Outside size: approximately **90 × 76 × 27 mm** when assembled.
- Best for: a hidden controller, short cable runs, or the smallest possible box.
- The USB-C hub, camera cable, power cable, and optional Ethernet cable remain outside the enclosure.



Controller Dock exterior

Controller Dock

Choose the Dock when the UNO Q and Arduino hub should move and mount as one unit. The hub sits in an open service bay, so its USB, USB-C power, and Ethernet connections remain reachable.

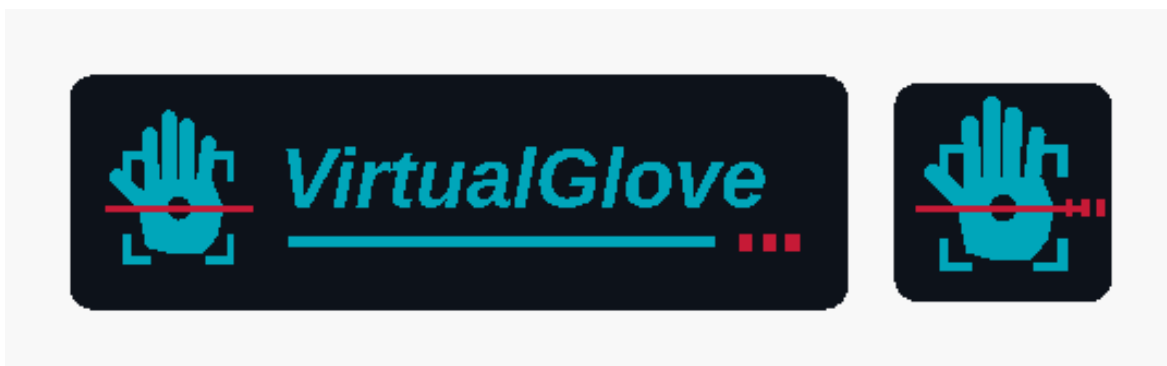
- Outside size: approximately **148 × 104 × 31 mm** when assembled.
- Best for: a neat arcade-panel installation or a Controller given as a complete unit.
- The hub remains removable; it is not trapped inside a sealed hot box.

What is included

The source and ready-to-print files are in [hardware/enclosures/](#).

Part	Purpose
UNO base and lid	Compact UNO Q enclosure
Dock base and lid	Combined UNO Q and hub enclosure
Matrix bezel	Cyan frame around the visible Matrix
Wordmark badge	Small lid badge for either enclosure
Full-logo layers	Optional dark, cyan, and red display plaque
Hand/target layers	Optional dark, cyan, and red emblem
USB-C coupon	Checks the UNO Q plug opening before a long print
Hub coupon	Checks the Arduino hub cradle before a long print

The full-logo and hand/target pieces are printable adaptations of the project artwork. They preserve the italic wordmark, hand, target, cyan linework, and red accents while removing screen-only glow and details smaller than a dependable 0.4 mm nozzle can reproduce.



Print-safe full logo and hand-target emblem

Before printing

You will need:

- an Arduino UNO Q;
- the Arduino USB-C Hub (8 in 1) for either design;
- four M3 × 8 mm screws for the board;
- four M3 × 8-12 mm screws for the case;
- four M3 heat-set inserts with an outside diameter near 4.0-4.2 mm;
- four adhesive rubber feet; and
- optional 1 mm foam or TPU strips beneath the hub.

The design follows Arduino's published **68.58 × 53.34 mm** UNO Q outline and four mounting holes. The hub envelope is **119 × 27.8 × 16 mm** with a fixed 175 mm cable. Arduino does not publish every plug-body and overmould dimension, so the coupons are part of the build process rather than an optional extra.

Reference drawings:

- [Arduino UNO Q datasheet](#)
- [Arduino UNO Q STEP model](#)
- [Arduino USB-C Hub datasheet](#)

First check-in: print the coupons

Print these before the enclosure:

1. [virtualglove-usb-c-fit-coupon.stl](#)
2. [virtualglove-hub-fit-coupon.stl](#)

The USB-C plug should enter without scraping and should not have enough side play to pull hard against the UNO Q socket. The hub should sit in its channel without bowing the coupon or rattling loosely.

Record the printer, material, nozzle, slicer profile, and any dimensional change you make. In [virtualglove-controller.scad](#), [fit](#) controls ordinary mating clearance and [hub_fit](#) controls the hub cradle. Do not scale the entire model to fix one opening; change the relevant clearance instead.

Kobra 3 PLA starting profile

- 0.4 mm nozzle
- 0.20 mm layer height
- four walls
- five top and bottom layers
- 20-25% gyroid infill
- supports off initially
- brim only if the base corners lift
- base and lid in the supplied orientation

The lid's visible face lies on the bed for a clean finish. PLA is suitable inside a normal arcade cabinet. Do not leave the Controller in a hot car or another high-temperature location, and never cover its ventilation slots.

Printing the branding

The smallest lid badge is a one-piece relief and can be printed in one colour. For the closest match to the VirtualGlove artwork, use charcoal or black for the backing, cyan for the hand and wordmark, and red for the target beam and terminal accents.

The colour layers are separate STL files. With one ACE Pro, assign each layer its colour in the slicer and assemble them on the same plate, or print one colour at a time and glue the layers to the backing. Separate accents avoid wasting material on colour changes through an entire enclosure.

Use a small amount of thin double-sided adhesive or plastic-safe glue. Dry-fit the pieces first; the detailed layers are intentionally shallow and should not be forced.

Assemble the UNO Q Case

1. Heat the four inserts into the corner bosses. Keep them square and stop when they are flush; excess heat can soften the boss.
2. Fasten the UNO Q to the four standoffs.
3. Connect the Arduino hub through the broad USB-C opening.
4. Dry-fit the lid and confirm the plug does not press on the socket.
5. Add the Matrix bezel and preferred badge.
6. Fasten the lid and add the rubber feet.

Only the UNO Q USB-C port is intentionally exposed. Do not force a thick plug through the opening; adjust and reprint the coupon if necessary.

Assemble the Controller Dock

1. Install the inserts and UNO Q as described above.
2. Add thin foam or TPU strips to the hub cradle if desired.
3. Route the hub's captive cable through the internal channel and connect it to the UNO Q.
4. Seat the hub in the open rear bay.
5. Confirm both long port faces, the Ethernet end, and the captive lead are free.
6. Connect the camera to a USB-A 3.0 port, power to the hub's USB-C PD input, and optional Ethernet.
7. Fit the lid, Matrix bezel, and branding, then add the rubber feet.

Second check-in: inspect before power

Before switching on the Controller, confirm:

- the board is flat and no standoff is twisting it;
- screw heads and inserts cannot touch components or solder joints;
- the hub and USB-C cable are not pushing sideways on the UNO Q socket;
- the Matrix window and every ventilation opening are clear;
- no loose screw, wire strand, or printed debris remains inside; and
- the lid closes without pressure on the board, hub, or cables.

Power the assembled Controller for 30 minutes with the camera active. Check that the case remains comfortable to touch, the camera stays connected, and the Matrix and Dashboard behave normally. A warm board is expected; a softening PLA case, repeated USB disconnect, or unusually hot enclosure is not.

Make a revision

Open [virtualglove-controller.scad](#) in OpenSCAD. The dimensions and clearances that are most likely to need adjustment are grouped at the top of the file. Choose a part from the [part](#) selector, preview it, and export it directly, or regenerate the complete checked-in STL set with:

```
BASH
hardware/enclosures/export-stl.sh
```

Keep the source file, STL files, and this guide together in a check-in. Note the printer and measured fit when changing a hardware dimension so the next person can tell whether the change is universal or printer-specific.

Safety and limits

- Disconnect power before opening the case.
- Handle the UNO Q with normal electrostatic-discharge care.
- Never force the board, hub, plug, insert, or lid into place.
- Keep metal fasteners and decorative foil away from antennas and electronics.
- Keep airflow paths open and inspect PLA periodically in a warm cabinet.
- Treat this as a first-fit printable design until it has been measured against the exact hub, plugs, inserts, and printer that will be used.

The enclosure protects and organizes the Controller; it does not make the UNO Q or hub waterproof, impact-rated, or suitable for outdoor use.